TABLE 1
BUTTERFLY IQ PROBE SPECIFICATIONS

Item	Probe Specification
Probe dimension	85×56×35mm
Probe weight	313 grams
Power	Battery (rechargeable)
Battery life	2 hours for continuous scanning
Display	Variable
Min/Max scan depth	1cm min / 30cm max
Ultrasound chip	Integrated CMOS chip
Transducer	9000-element CMUT
Frequency range	1-10MHz
Cable length	Lighting: 1.25 metres USB-C: 2 metrev

A study showed that first-year medical students using ultrasound can diagnose cardiac abnormalities better than expert cardiologists using stethoscopes. There are many other use cases where ultrasound scanning is more powerful than the clinical exam, for example, from detecting an aneurysm to rapidly diagnosing fluid in the abdomen.

Similarly, using ultrasound to visualise a blood vessel when placing a catheter has shown significant benefit relative to doing it blind, which is the present clinical standard in some cases. With time, this technology might allow the patients to do their own ultrasound scanning. Butterfly Network is making an early start in this regard.

TABLE 2
PIEZOELECTRIC CRYSTAL BASED ULTRASOUND SCANNING VS CHIP BASED ULTRASOUND SCANNING

Piezoelectric Based Ultrasound Scanning	Ultrasound-on-a Chip Based Ultrasound Scanning
Piezoelectric based crystal produces ultrasound	Chip produces ultrasound
Costs around $8,000	Butterfly network probe iQ costs $1,999 (at release)
For each application a separate probe is required	All applications can be done with a single probe
Needs more storage space	Needs less storage space
Domestic scanning is not possible	Domestic scanning is possible
Image clarity is high	Image clarity is less
More reliable	Less reliable

Jonathan Rothberg, an entrepreneur in the biomedical industry, is behind the development of Butterfly iQ probe. This iQ probe is FDA-approved for 13 different applications with a single probe, which is the maximum ever for a single ultrasound device.

Just one high-bandwidth tool can scan the abdomen, the blood vessels, the thyroid gland, and various other systems, in addition to guiding medical procedures. Doing it all with a single probe is hugely beneficial for the physicians and reduces scanning space. Although this portability might sound like mere convenience, its mass production in future might replace the stethoscope.

Butterfly iQ probe works with Apple and Android devices using the Butterfly iQ app. Butterfly iQ probe specifications are given in Table 1 above.

In conclusion, just like a domestic blood sugar tester, it can make home scanning possible. It can reduce scanning cost to a great extent and make medical imaging technology easy and available everywhere. It can revolutionise medical imaging in hospitals and clinics. It can change the game in global health and soon become a consumer product that will be as easily available as the household thermometer. EFY